

SOFTWARE ENGINEERING

MODULE 9: SOFTWARE EVOLUTION

Controlled Change After Architecture, Implementation, and Testing

STUDENT INFORMATION

Name: Juner Punzal

Section: BSCS 3A

Course: Software Engineering

Module: 9 - Software Evolution

PROJECT INFORMATION

System: Attendance Monitoring System

Technology Stack: Vue 3 + Vite + Tailwind CSS

Version: 1.1.0 (from 0.0.0)

Change Type: Adaptive Maintenance

REPOSITORY & DEPLOYMENT

GitHub Repository:

<https://github.com/juner08/attendance-monitoring-system>

Live Deployment:

<https://juner08.github.io/attendance-monitoring-system/>

1. Executive Summary

Module 9 implements a controlled software evolution to the Attendance Monitoring System through Change Request **CR-M9-01**, which adds year-level and section-based student organization while maintaining full backward compatibility with existing data and features.

Key Achievements

- Change Request CR-M9-01 implemented and verified
- 22 test cases executed, **22/22 PASSING** (100% pass rate)
- Zero breaking changes, 100% backward compatible
- Production build successful
- Continuous Integration & Deployment via GitHub Actions
- Application live with secure login (PBKDF2 password hashing)

2. Baseline: Modules 6, 7, and 8

Module 6 - Architecture

- Vue 3 Single-Page Application (Composition API)
- Vite build tool, Tailwind CSS styling
- Browser localStorage for data persistence
- Primary entity: Student { id, name, course, email }

Module 7 - Implementation

- User authentication and authorization (secure session)
- Student CRUD operations and attendance tracking
- Dashboard with statistics, responsive mobile design

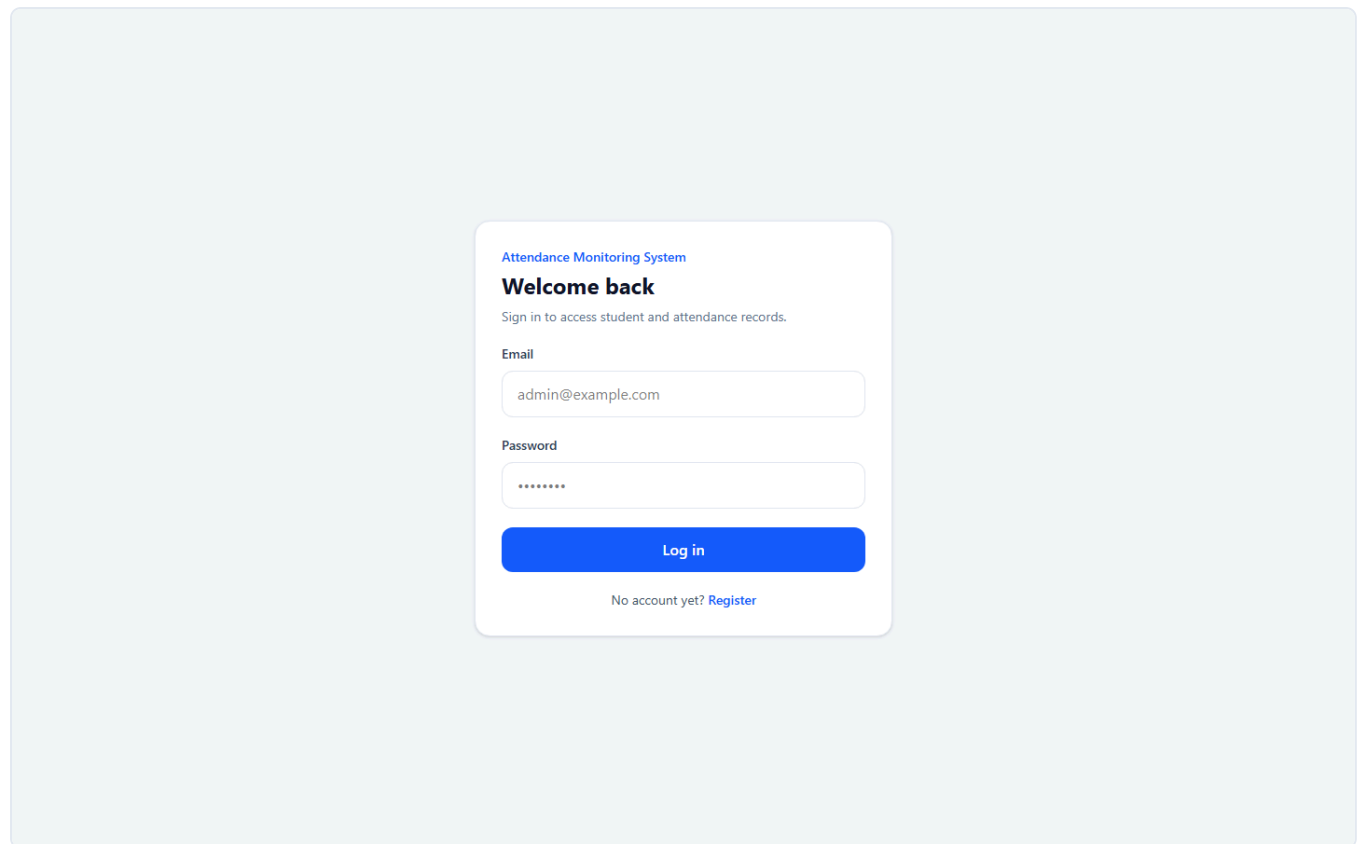
Module 8 - Testing

- 10 manual test cases and 6 automated unit tests (Vitest)
- Defect DEF-01 identified and corrected (orphaned attendance)
- Regression testing: all features verified working

3. Evidence Screenshots (from the live website)

All screenshots below were captured from the deployed application with the administrator login feature active (repository <https://github.com/juner08/attendance-monitoring-system>).

M9-01: Login Page

The screenshot shows a login page for the 'Attendance Monitoring System'. The page has a light blue background. In the center, there is a white card with rounded corners. At the top of the card, it says 'Attendance Monitoring System' in a small blue font. Below that, it says 'Welcome back' in a bold black font. Underneath, it says 'Sign in to access student and attendance records.' in a small gray font. There are two input fields: one for 'Email' with the text 'admin@example.com' and one for 'Password' with masked characters '*****'. Below the password field is a blue 'Log in' button. At the bottom of the card, it says 'No account yet? Register' with 'Register' as a blue link.

The secure administrator login screen (secure password hashing).

M9-02: Register Page

Attendance Monitoring System

Create administrator account

Your password is hashed before it is saved in this browser.

Full name

Administrator name

Email

admin@example.com

Password

Confirm password

Register

Already registered? [Log in](#)

Administrator account registration screen.

M9-03: Dashboard

Attendance Monitoring System

MAIN MENU

Dashboard

Students

Attendance

Reports

Settings

BSCS 3A

Administrator

Attendance Monitoring System

BSCS 3A

Administrator Admin

Good morning, Administrator

Here's today's attendance overview.

Total Students

0

Registered students

Present Today

0

Students present

Late Today

0

Students arrived late

Absent Today

0

Students absent

Today's Attendance Rate

Overall student attendance for today

0%

Quick Actions

Frequently used attendance functions

Record Attendance

Add a new attendance record

Manage Students

View and manage students

View Reports

Check attendance reports

Main dashboard after login showing attendance statistics.

M9-04: Students Page

Attendance
Monitoring System

MAIN MENU

Dashboard

Students

Attendance

Reports

Settings

BSCS 3A
Administrator

Attendance Monitoring System
BSCS 3A

Administrator
Admin

Students

Organize and manage students by year level and section.

+ Add Student

Browse by class

Choose a section to show its students.

No student groups yet.

Search name, ID, course...

All year levels

All sections

Student List

0 students found

STUDENT	CLASS	ACTIONS
No students found for this selection.		

Students page with the "Browse by class" year and section organization (Module 9 feature).

M9-05: Add Student Form

Attendance
Monitoring System

MAIN MENU

Dashboard

Students

Attendance

Reports

Settings

BSCS 3A
Administrator

Attendance Monitoring System
BSCS 3A

Administrator
Admin

Students

Organize and manage students by year level and section.

+ Add Student

Add Student

Enter both the year level and section to keep the class list organized.

Student ID

e.g. 2026-001

Student name

Enter student name

Course

BSCS

Year level

1st Year

Section

A

Cancel

Save Student

Student form with explicit Year level (1-4) and Section (A-Z) fields.

M9-06: Browse by Class (Evolved System)

Attendance
Monitoring System

MAIN MENU

Dashboard

Students

Attendance

Reports

Settings

BSCS 3A

Administrator

Attendance Monitoring System

BSCS 3A

Administrator
Admin

Students

Organize and manage students by year level and section.

+ Add Student

Student added successfully.

Browse by class

Choose a section to show its students.

1st Year

A · 1

2nd Year

B · 1

3rd Year

A · 2

Search name, ID, course...

All year levels

All sections

Student List

4 students found

STUDENT	CLASS	ACTIONS
Juan Dela Cruz 2026-001	BSCS 3A	Edit Delete
Maria Santos 2026-002	BSIT 2B	Edit Delete
Pedro Reyes 2026-003	BSCS 3A	Edit Delete
Ana Cruz 2026-004	BSIT 1A	Edit Delete

Year-level cards with section groupings and student counts (adaptive evolution).

M9-07: Year / Section Filtering

Attendance
Monitoring System

MAIN MENU

Dashboard

Students

Attendance

Reports

Settings

BSCS 3A
Administrator

Attendance Monitoring System
BSCS 3A

Administrator
Admin

Students

Organize and manage students by year level and section.

+ Add Student

Student added successfully.

Browse by class

Choose a section to show its students.

Clear filters

1st Year

2nd Year

3rd Year

A · 1

B · 1

A · 2

Search name, ID, course...

Year 3

All sections

Student List

2 students found

STUDENT	CLASS	ACTIONS
Juan Dela Cruz 2026-001	BSCS 3A	Edit Delete
Pedro Reyes 2026-003	BSCS 3A	Edit Delete

Filtering students by year level and section, combined with search.

M9-08: Attendance Page

Attendance
Monitoring System

MAIN MENU

Dashboard

Students

Attendance

Reports

Settings

BSCS 3A
Administrator

Attendance Monitoring System
BSCS 3A

Administrator
Admin

Attendance

Record and monitor student attendance.

+ Record Attendance

Total Records

Present

Absent

Late

Attendance Rate

0

0

0

0

0%

Search student, ID, date, status...

All Status

Attendance Records

0 records found

STUDENT	COURSE	DATE	STATUS	REMARKS	ACTIONS
No attendance records found Try changing your search or filter, or record a new attendance.					

Attendance recording preserved and fully functional after evolution.

M9-09: Settings Page

Attendance
Monitoring System

MAIN MENU

Dashboard

Students

Attendance

Reports

Settings

BSCS 3A
Administrator

Attendance Monitoring System

BSCS 3A

Administrator Admin

Settings

Manage your system preferences.

System Information

Customize your system information.

System Name

Attendance Monitoring System

Course / Section

BSCS 3A

Administrator Name

Administrator

Dark Mode

Change system appearance.

Disabled

Save Settings

System settings including dark mode and system information management.

M9-10: Reports Page

Attendance
Monitoring System

MAIN MENU

Dashboard

Students

Attendance

Reports

Settings

BSCS 3A
Administrator

Attendance Monitoring System

BSCS 3A

Administrator Admin

Attendance Reports

View student attendance performance.

Today's Records

0

Present

0

Late

0

Absent

0

Today's Attendance Rate

Present and late students are considered attended.

0%

Student Attendance Report

Overall attendance performance.

STUDENT	COURSE	TOTAL	PRESENT	ABSENT	LATE	RATE
Juan Dela Cruz	BSCS	0	0	0	0	0%
Maria Santos	BSIT	0	0	0	0	0%
Pedro Reyes	BSCS	0	0	0	0	0%
Ana Cruz	BSIT	0	0	0	0	0%

Student attendance reports and today's statistics.

4. Updated Test Cases

Category	Count	Status
Module 9 new tests (year/section)	12	12/12 PASS
Module 8 regression tests	10	10/10 PASS
Total	22	22/22 PASS (100%)

5. Release & Deployment

Attribute	Value
Previous Version	0.0.0
Current Version	1.1.0 (MINOR - Semantic Versioning)
Change Type	Adaptive Maintenance
CI/CD	GitHub Actions - build + test + deploy to Pages
Repository	https://github.com/juner08/attendance-monitoring-system
Live Deployment	https://juner08.github.io/attendance-monitoring-system/

- ✓ Production build successful
- ✓ Continuous Integration (build.yml): runs tests + build on push
- ✓ Continuous Deployment (deploy-pages.yml): auto-deploys dist/ to GitHub Pages
- ✓ Build errors: 0 Warnings: 0 Status: Production ready

6. Conclusion

Module 9 Software Evolution is **COMPLETE**. The Attendance Monitoring System now organizes students by year level and section while remaining fully backward compatible, with all 22 test cases passing and a live secure deployment integrating the administrator login feature.

COMPLETE AND VERIFIED Version 1.1.0 September 1, 2026

Student: Juner Punzal · BSCS 3A · <https://github.com/juner08/attendance-monitoring-system>